Cyclistic Bike-Share Analysis

By Jose Nodarse

Comparing casual and annual member trip patterns using 2019 historical data

Business Task:

Analyze recorded bike trips during calendar year 2019 to identify differences between casual and annual member trips in duration, departure hour, day of the week, and month. Use these findings to propose research and marketing experiments that could support the conversion of casual riders into annual members.

Stakeholders and Expectations:

Stakeholder	Expectations for the analysis
Cyclistic Executive Team	Business implications, limitations and recommendations
Lily Moreno, Marketing Manager	A clear marketing direction supported by reproducible analysis
Marketing Analytics Team	Useful and evidence-backed differences between annual members and casual users

Analysis Focus: Questions to Answer

1. How do typical trip durations differ between casual riders and annual member trips?
2. How do departure hour patterns differ between the groups?
3. How does weekday and weekend activity differ?
4. How does trip activity vary by month?
5. What share of recorded trips belong to each group?
6. How sensitive are duration summaries to unusually long records.

Source Breakdown

This analysis uses four quarterly Divvy trip datasets covering trips beginning from January through December 2019. Divvy data represents the fictional Cyclistic company for this certificate case study. The files contain trip identifiers, start and end timestamps, reported durations, station details, user types, and demographic fields. The combined dataset contains 3,818,004 recorded trips. “Customer” records are labeled “Casual,” and “Subscriber” records are labeled “Member” throughout the analysis.

The year 2019 was selected to provide a consistent historical full-year comparison. It should not be described as the latest available year, and its findings require validation before informing current campaigns.

Credibility:

Source and consistency: The quarterly files have compatible trip-level fields after standardizing the Q2 column names. Validation found no duplicate trip IDs across the combined dataset and no exact duplicate rows within the quarterly files.

Completeness: Gender is missing for 559,206 records, and birth year is missing for 538,751 records. These fields were not used because demographic analysis is outside the study's scope and missingness could bias demographic comparisons. Their missingness does not, by itself, invalidate the trip-timing analysis.

Historical relevance: The analysis describes 2019 activity. Changes in pricing, infrastructure, customer behavior, and service operations may limit its relevance to current decisions.

Interpretive limits: The records do not identify unique riders or directly record trip purpose, membership motivations, or responses to marketing.

Tools and Rationale for Use

Tool	How it was used	Why it fits
Python/pandas	Read CSVs, standardize, combine, validate, filter, and summarize trips	Supports reproducible results across multiple large CSV files
matplotlib	Create and export count and percentage charts	Allows for the use of findings to be made into clear visuals that can be exported
Jupyter notebooks	Organize code, explanations, outputs, and analytical decisions	Makes the workflow easier to review and reproduce.

Cleaning and Transformation Log

Step	Action and Reasoning
Preserve source data	Keep all CSV files and create separate working DataFrames
Standardize columns	Explicitly rename Q2 column names to match the other quarters
Combine quarters	Stack quarterly records and retain a source file
Covert data types	Parse timestamps and convert duration times from text to numeric values
Validate Records	Check for duplicates, missing values, timestamp coverage, user types, and duration inconsistencies.

Set scope	Select eligible trips beginning during calendar year 2019
Make analysis fields	Derive duration in minutes, departure hour, weekday, month, and weekday/weekend category.
Sensitivity analysis	Compare baseline summaries with summaries restricted to trips lasting at most 24 hours
Export	Produce reporting tables and charts from the analysis to export.

Quality Checks Observed

Check	Result
Total Records	3,818,004 rows
Data Range	January 1st - December 31, 2019
Duplicate trip IDs across all quarters	0
Exact duplicate rows within quarterly files	0
Missing/unparseable start or end timestamps	0
Missing, unparseable, or nonpositive durations	0
Reported durations over 24 hours	1,848
Reported durations differ from timestamp elapsed time by more than one second*	103
Null cells	Gender: 559,206 Birth year: 538,751

*This checks to see if there is a difference of more than one second between the reported duration and the calculated duration using end and start times.

Some trips beginning in 2019 ended in 2020. Inclusion is based on the start date. Extreme durations and differences between reported and timestamp-derived durations were flagged for investigation. The baseline retains reported durations; a separate sensitivity analysis examines the influence of records exceeding 24 hours.

Analysis and Interpretation

Measure	Casual Rider	Annual Member
Recorded trips	880,637	2,937,367
Share of recorded trips	23.1%	76.8%
Mean duration	57.0 minutes	14.3
Median duration	25.8 minutes	9.8
Percentage of trips exceeding 20 minutes	63.42%	16.66%
Percentage of trips exceeding 60 minutes	16.52%	0.42%
Percentage of trips exceeding 24 hours	1,347	501

Further Analysis

Casual trips have a median duration of 25.8 minutes, compared with 9.8 minutes for member trips. This indicates a substantial difference in typical recorded trip duration.

Means exceed medians in both groups, indicating right-skewed duration distributions. Approximately 16.52% of casual trips exceed one hour, compared with 0.42% of member trips.

Extreme records influence the averages. Restricting the comparison to trips lasting at most 24 hours reduces the casual mean from 57.0 to 39.4 minutes and the member mean from 14.3 to 12.9 minutes. The medians change little. Therefore, medians are emphasized when describing typical trips.

Duration Summary	Casual	Member
Baseline mean	57.0 min	14.3 min
Mean for trips ≤24 hour	39.4 min	12.9 min

The 24 hour restriction is an exploratory sensitivity check. It is not meant to be used as a rule to invalidate trips.

Hourly Activity

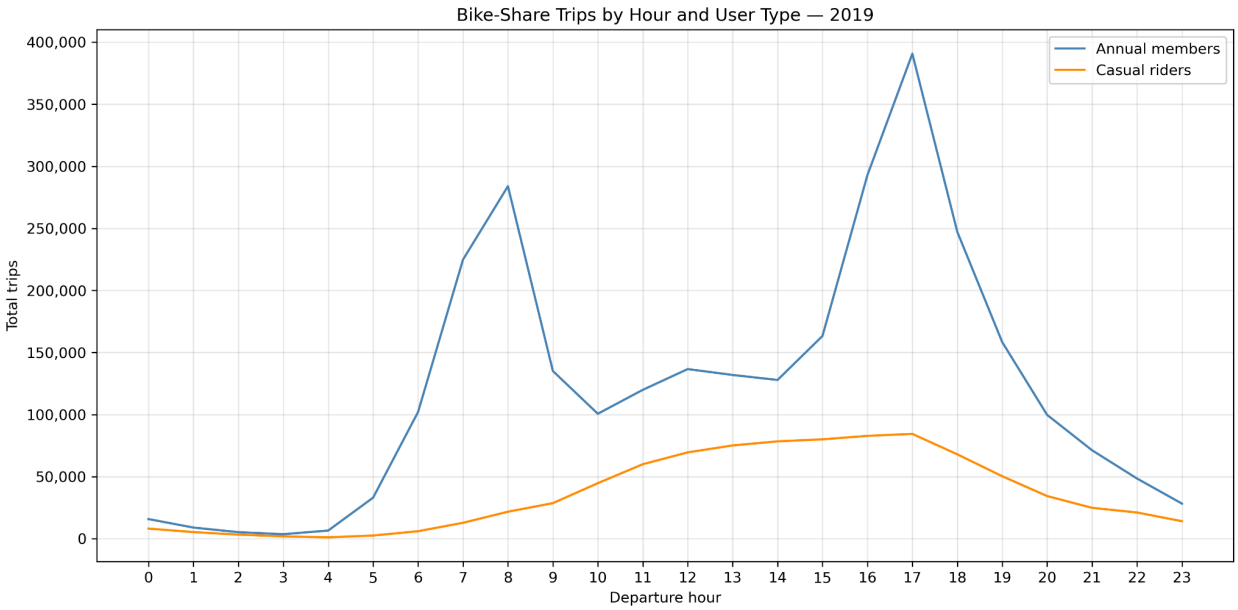


Figure 1: Hourly distribution of total recorded rides for casual riders and annual members.

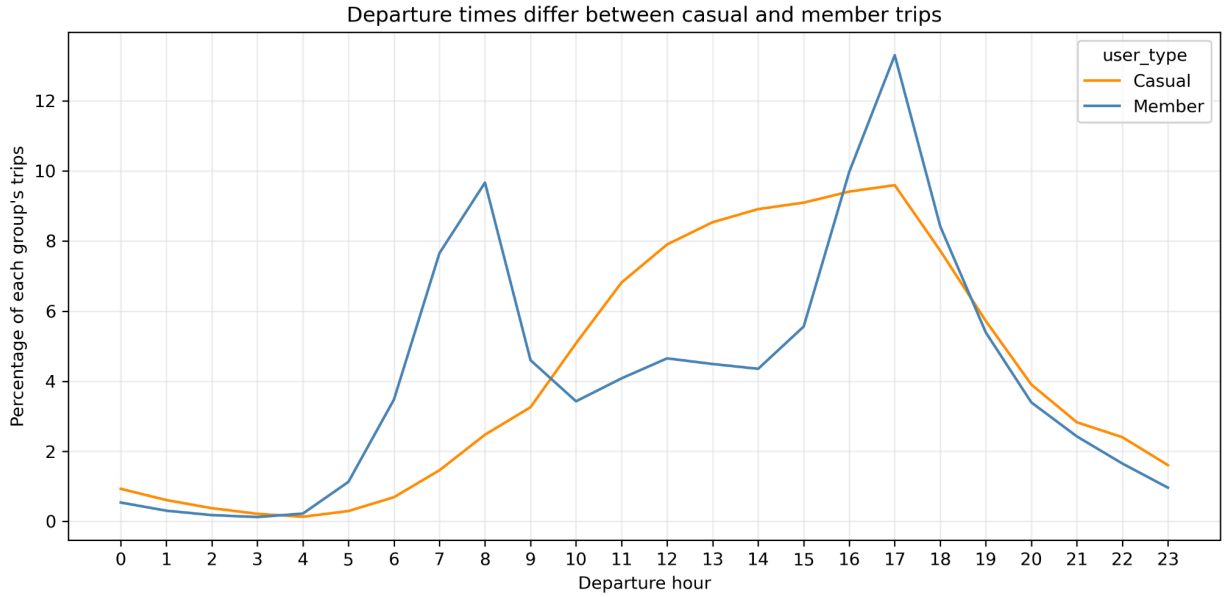


Figure 2: Percentage of trips per hour for casual riders and annual members

Member departures show pronounced morning and evening peaks, including 8 a.m. and 5 p.m. Casual departures have a broader afternoon concentration, with their highest hourly count at 5 p.m.

The member pattern is consistent with commuting, but trip purpose is not observed. Within-group percentages complement raw counts by showing how each group distributes its trips across the day despite their unequal totals.

Weekday and Weekend Activity

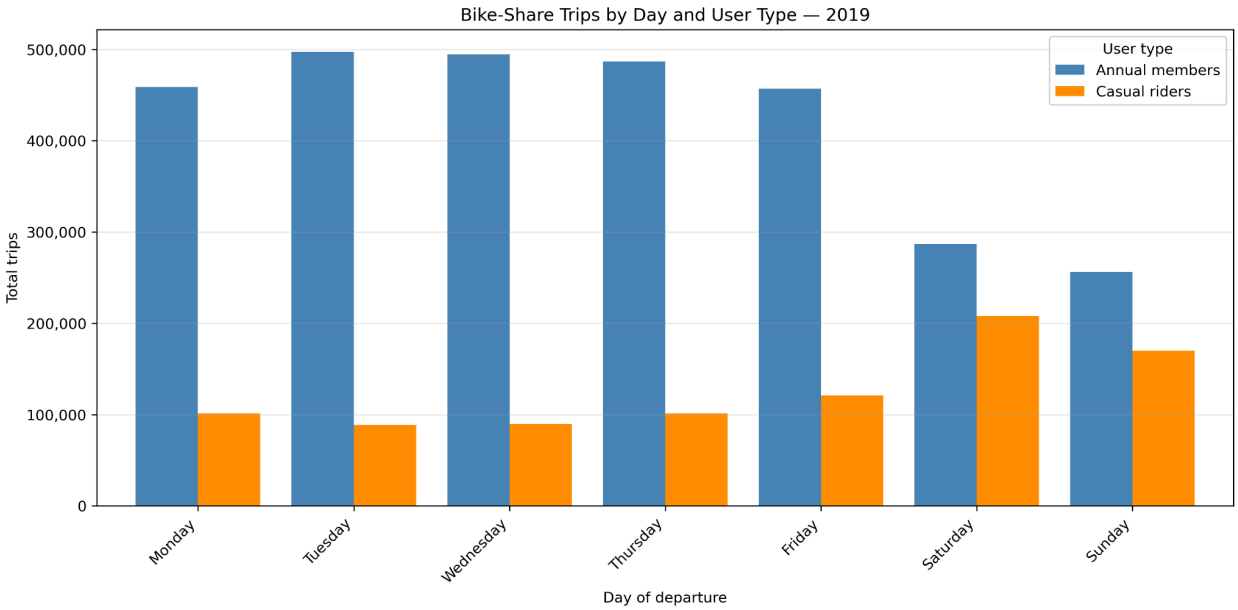


Figure 3: Total rides per day of the week

Weekdays account for 81.5% of member trips and 57.0% of casual trips. Weekends account for 18.5% and 43.0%, respectively.

Because weekdays and weekends contain different numbers of calendar days, daily averages provide a clearer comparison. Casual trips average approximately 1,925 per weekday and 3,637 per weekend day—about 1.89 times as many per weekend day. Member trips average approximately 9,172 per weekday and 5,225 per weekend day.

Monthly Activity

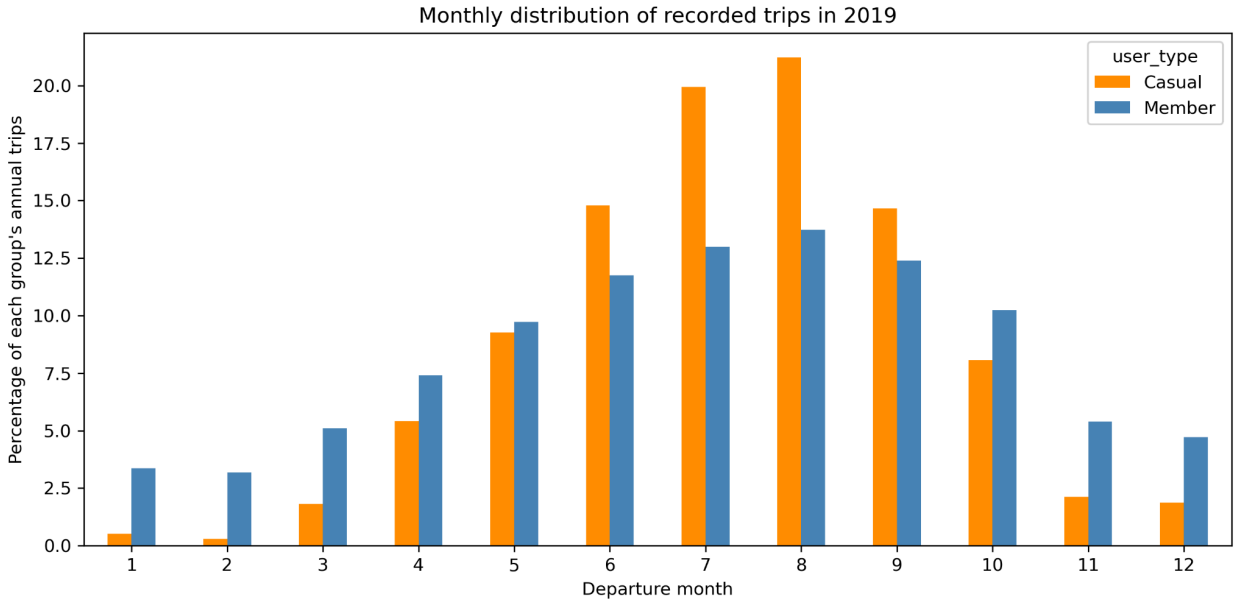


Figure 4: Percentage of each group's rides per month.

Both groups record their highest monthly trip counts in August. Monthly distributions show that activity varies substantially across the year. These results describe seasonality within 2019. The dataset alone cannot establish whether weather, tourism, service availability, or other factors caused the differences.

Final Analysis

Member trips account for approximately 76.9% of recorded activity and are typically shorter than casual trips. Their departure patterns are concentrated on weekdays and include pronounced morning and evening peaks. Casual trips are typically longer and have stronger weekend concentration. Casual departures show a broad afternoon peak. These patterns suggest different usage contexts, but they do not establish individual travel habits or reasons for riding. Commuting and recreational use remain hypotheses requiring additional evidence.

Recommendations

Recommendation	Proposed action	Evaluation
Understand membership barriers	Conduct an optional survey covering trip purpose, membership awareness, visitor/local status, and reasons for not subscribing(complicated process, pricing, etc.)	Response rate, respondent mid, and reported barriers
Test membership messaging	Compare a membership-value message against the existing experience for eligible casual riders	Conversion within a predefined period versus a randomized comparison group
Test Campaign timing	Use casual-trip timing patterns to propose campaign schedules, subject to available channels and eligibility	Incremental conversions and acquisition cost

These recommendations are proposed research and experiments, not proven conversion strategies. Implementation requires additional information about pricing, marketing permissions, targeting capabilities, and campaign costs.

Final Limitations

- Trips are observations; unique riders cannot be counted.
- Station endpoints do not reveal physical routes or individual repeat behavior.
- Trip purpose and membership motivations are unobserved.
- Extreme durations influence means, and duration discrepancies remain unresolved.
- The analysis is descriptive and does not establish causation.
- Results describe 2019 and require validation for current decisions.